



SAPTHAGIRI NPS
UNIVERSITY

UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

SCHOOL OF APPLIED SCIENCE

PROJECT SYNOPSIS REPORT

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COSMIC ZONE

1. INTRODUCTION

Space science has always been a subject of curiosity and inspiration for people of all ages, especially children. However, traditional learning methods such as textbooks and static content often fail to effectively explain complex space concepts like planetary motion, rocket launches, and real-time scientific discoveries. At the same time, access to reliable and simplified daily science and space-related news is limited for students and general users.

The Interactive Space Learning & Science News System is a web-based application designed to make space education engaging, visual, and interactive. The system provides a 3D interactive solar system that allows users, especially children, to explore planets visually without complex textual explanations. Each planet can be viewed in 3D, helping learners understand size, position, and movement intuitively.

In addition, the system includes a rocket launch simulation module, where users can select parameters such as rocket type, fuel capacity, and payload to observe simulated outcomes like launch success, distance traveled, or mission failure through animations or videos.

The project also integrates a Science News Module that fetches real-time daily space and science news using APIs. Users can view short summaries along with direct links to original articles, ensuring access to authentic and up-to-date information.

This system bridges the gap between education and technology by combining visualization, simulation, and real-world scientific updates into a single platform.

2. OBJECTIVE

- To provide an interactive and engaging platform for learning space science concepts.
- To help children understand the solar system using 3D visual representation.
- To simulate rocket launch scenarios based on user-selected parameters.
- To encourage learning through quizzes, MCQs, and badge-based rewards.
- To provide daily updated science and space-related news using APIs.
- To make science learning accessible for both kids and adults.
- To develop a responsive and user-friendly web application using modern technologies.

3. SCOPE OF THE PROJECT

- Students and Kids: Learn about planets, space, and rockets through interactive visuals.
- Teachers and Educators: Use the platform as a teaching aid for space science topics.
- General Users: Stay updated with daily science and space news.
- Educational Institutions: Use as a supplementary digital learning tool.
- Future Enhancements:
 - AR/VR-based space exploration
 - Live space mission tracking (ISRO/NASA)
 - AI-based personalized learning paths
 - Multiplayer quizzes and competitions

4. MODULES OF THE PROJECT

4.1 User Module

- User-friendly interface for navigation.
- Access to space learning, rocket simulation, and science news.
- Quiz participation and badge earning system.

4.2 Interactive Solar System Module

- 3D visualization of planets.
- Click-based interaction to explore each planet.
- Smooth animations for planetary movement.
- Designed mainly for kids with minimal text.

4.3 Rocket Launch Simulation Module

- User selects rocket parameters (fuel, payload, type).
- System simulates launch outcome.
- Displays results using animation or video.
- Educational explanation of success or failure.

4.4 Quiz & Learning Assessment Module

- MCQ-based quizzes on space topics.
- Instant feedback on answers.
- Badge and score system to motivate learning.

4.5 Science News Module

- Fetches daily space and science news using APIs.
- Displays short summaries.
- Provides external links to original news sources.
- Updated automatically every day.

4.6 Backend Module

- Handles API integration and data management.
- Manages quiz data, user progress, and simulation logic.
- Secure and scalable server-side logic.

4.7 Frontend Module

- Interactive UI with animations and visuals.
- Responsive design for mobile, tablet, and desktop.
- Clean layout suitable for kids and adults.

5. TECHNOLOGIES USING

Frontend:

- React.js
- JavaScript
- Three.js (for 3D visualization)
- HTML, CSS

Backend:

- Node.js

- Express.js

Database:

- MongoDB

APIs:

- Space & Science News APIs

Tools:

- VS Code
- GitHub
- Postman

6. EXPECTED OUTCOME

- A fully functional interactive space learning platform.
- Improved understanding of solar system concepts through 3D visuals.
- Engaging rocket launch simulations for conceptual learning.
- Increased interest in science among children and students.
- Real-time access to daily science and space news.
- A scalable and extensible educational web application.